TRANSLATING BUSINESS QUESTIONS INTO DATA-DRIVEN INSIGHT

Organisation: NHS

THE BUSINESS PROBLEM

The NHS must decide to either expand capacity or optimise already available resources. Two questions this analysis answers:



Was there adequate staff and capacity in the networks?



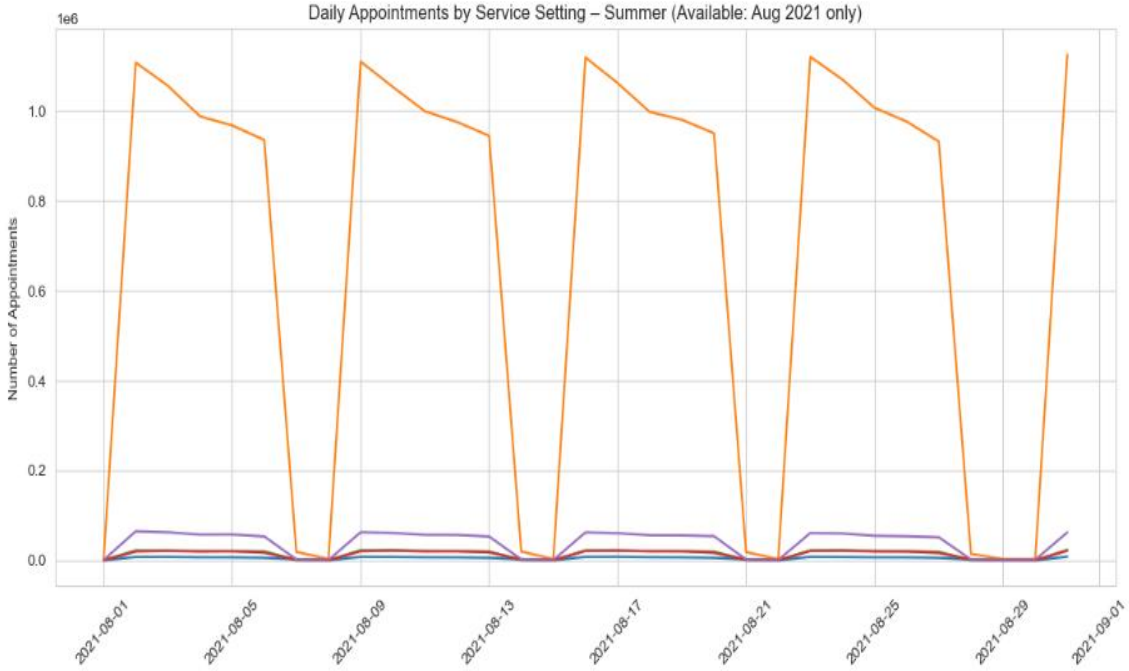
What was the actual utilisation of resources?

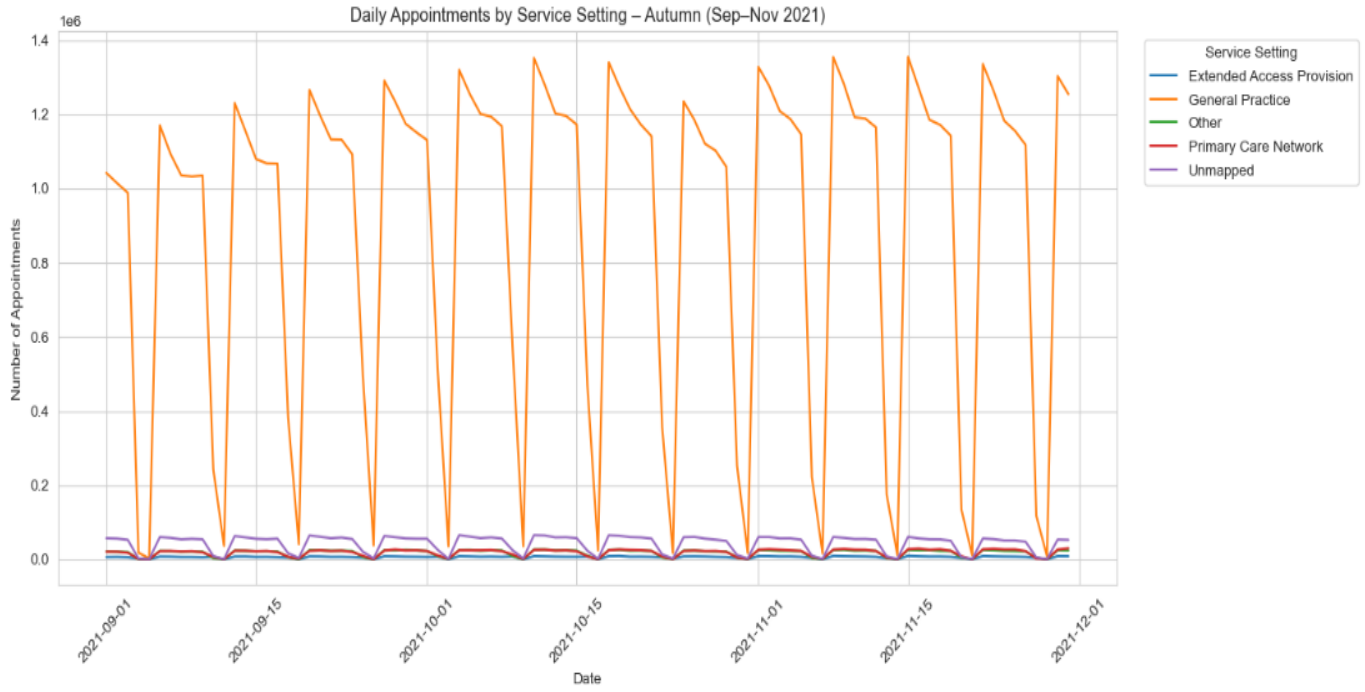
FOUNDATIONAL DATA FOR THE ANALYSIS

Appointment durations: how long each GP appointment lasted, by location and date (Dec 2021 – Jun 2022)

Regional appointment details: staff type, whether patients attended, how the appointment was booked, and how far ahead it was booked (Jan 2020 – Jun 2022)

National service categories: what kind of appointment it was and which part of the service delivered it, by location and date (Aug 2021 – Jun 2022)

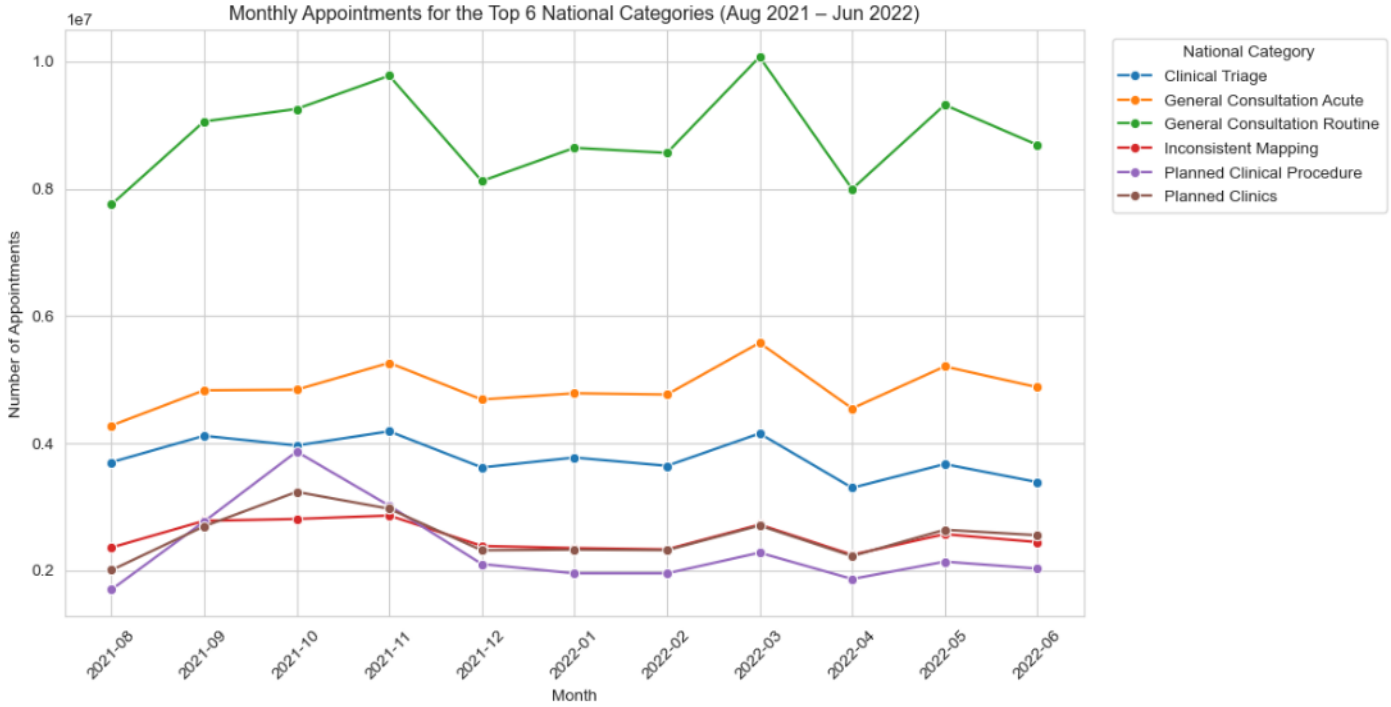


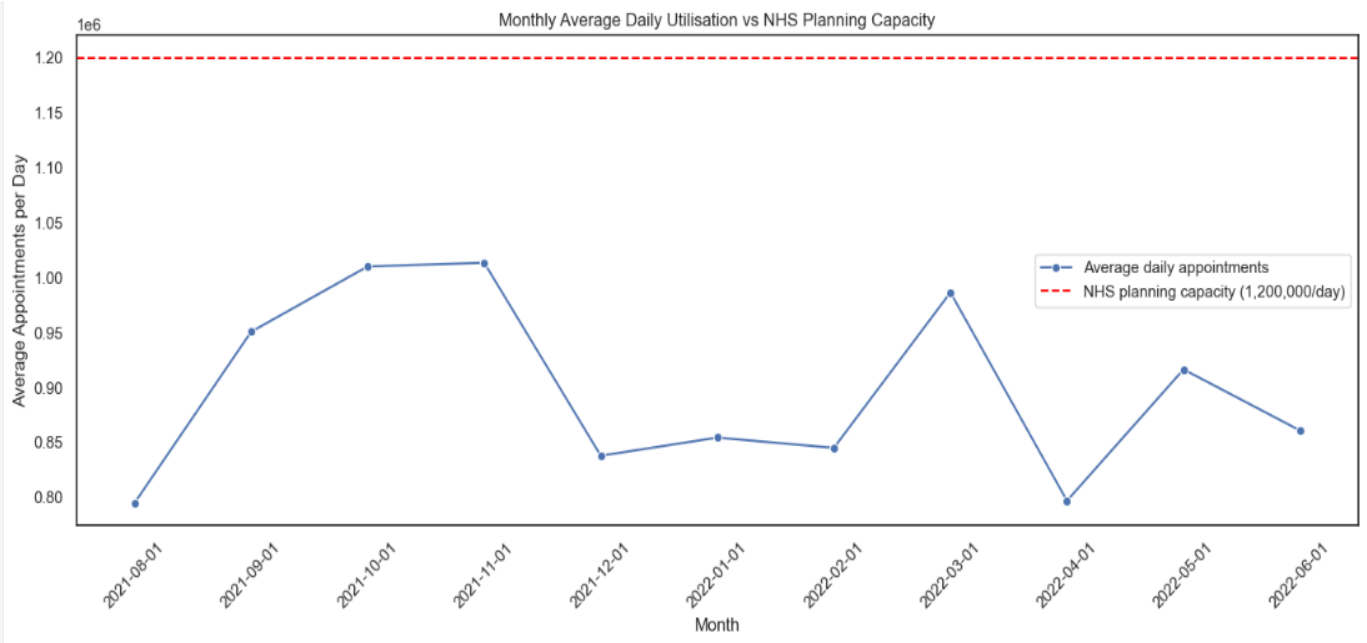


FINDING 1: THE WEEKLY CYCLE

- Appointments follow a sharp weekly pattern and are near zero on weekends
- General Practice alone reaches about 1.3M+ on peak weekdays in October - November
- This shifts the capacity conversation

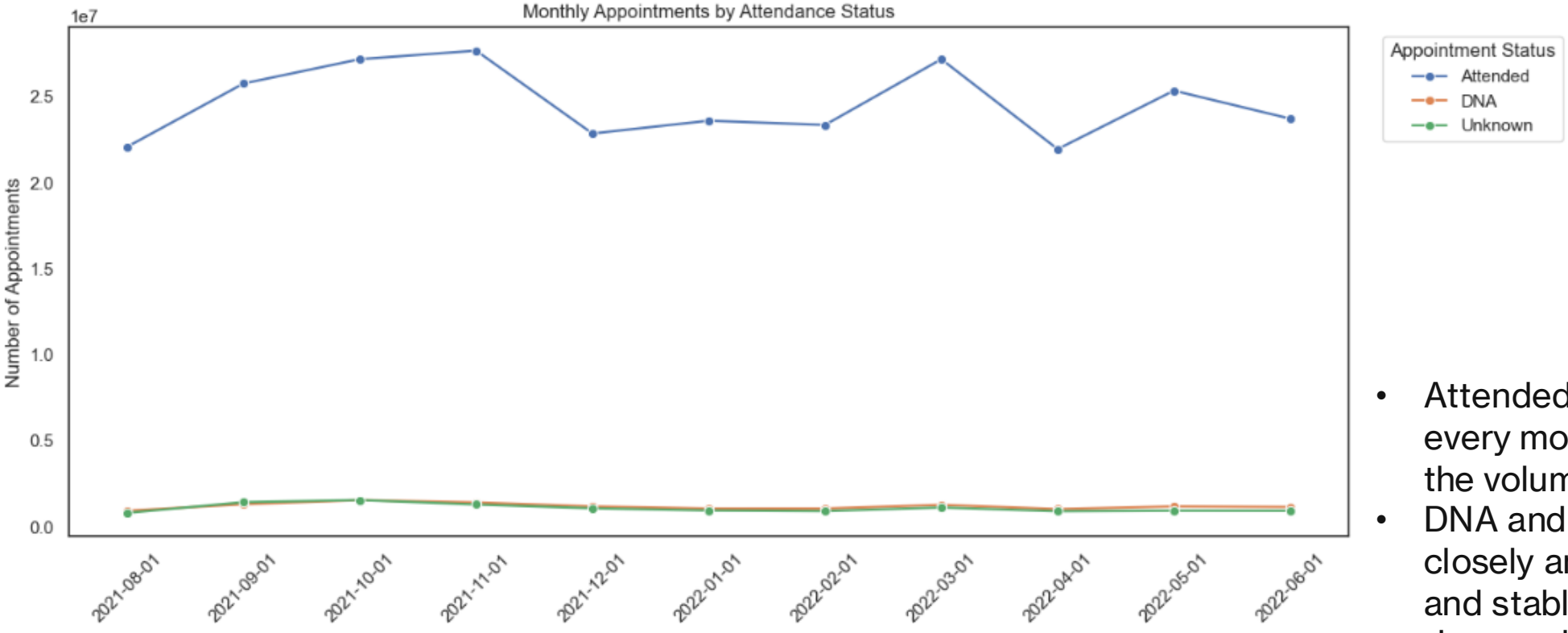
FINDING 2: A PREDICTABLE SEASONAL PEAK



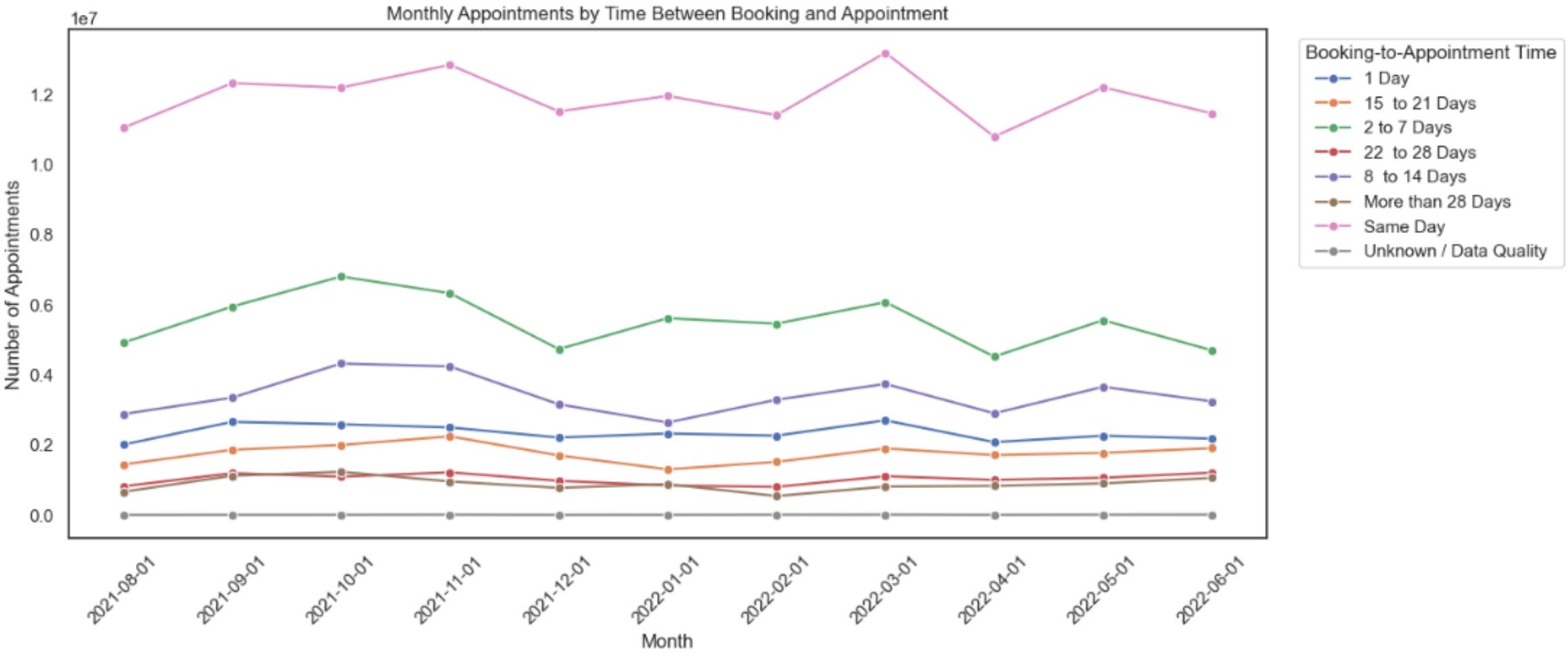


- October - November and March consistently peak across every category tested.
- December dips as well, but this is likely a holiday effect.

FINDING 3: WHAT ATTENDANCE DATA ACTUALLY SHOWS



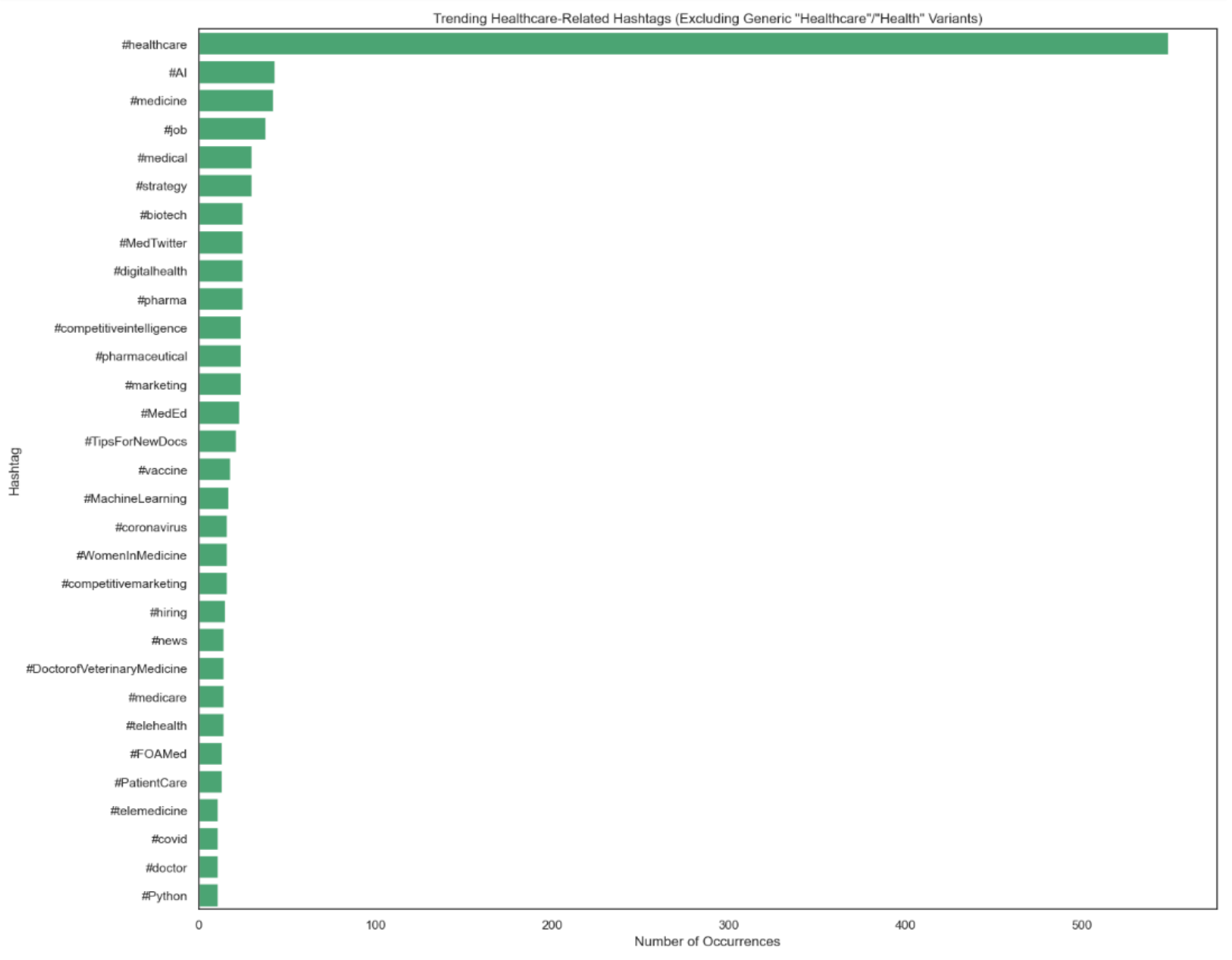
- Attended appointments dominate every month, running at roughly 20x the volume of DNA or Unknown.
- DNA and Unknown track each other closely and stay proportionally small and stable, even as total volume rises and falls.
- Same Oct - Nov peak, Dec dip, Mar peak pattern appears here too; attendance moves with overall demand, not against it



FINDING 4: BOOKING-TO-APPOINTMENT TIME

- "Same Day" is the largest booking-to-appointment time category every month, by a wide margin over every other band.
- It scales with overall demand, rising during the Oct - Nov and March peaks without shrinking as a share of the total.

Does Twitter/X Data Add Value?



- Useful for surfacing public topics of conversation
- Real limitations: no timestamps, small sample, reflects vocal users, not the general patient population.
- Data ethics: publicly available but using it to inform NHS decisions still requires care around consent, representativeness, and misuse of identifiable content.

ANALYTICAL RECOMMENDATIONS

1

Confirm the exact combined daily total across all service settings on peak days, against the 1.2M benchmark

2

Extend to a second year of data to confirm seasonality repeats.

3

Link booking wait times directly to missed-appointment rates.

4

Resolve the "Unknown"/"Unmapped" data gaps at the source.

CONCLUSION

Was capacity adequate?	Monthly averages say yes as utilisation, but during peak periods, the benchmark was crossed.
	But appointments cluster almost entirely on weekdays, not spread across all 30 days, so a monthly average understates true weekday pressure
What was actual utilisation?	“General Practice” and “Attended” appointments dominate as most capacity is converting into delivered care and not going to waste.
	A third of key records are still "Unknown" or "Unmapped”. This caps how precisely utilisation can currently be quantified.